

Ajay Suthaharan

647-554-9642 | asuth075@uottawa.ca | linkedin.com/in/ajaysuthaharan/ | github.com/Ajay757

EXPERIENCE

Machine Learning Research Engineer Co-op

Jan. 2026 – Apr. 2026

NRCan

Ottawa, ON, Canada

- Analyzed and compared foundation model architectures for remote sensing and Earth observation.
- Evaluated spatiotemporal ML model families to inform architectural decisions for large-scale geospatial data.
- Collaborated with researchers to develop a public taxonomy categorizing foundational models for remote sensing.
- Authored a research paper to standardize architectural terminology and enable clearer model comparison and adoption.

Software Developer Internship

May. 2024 – Aug. 2024

Driftstack

Greater Toronto, ON

- Revamped and optimized the website UI with React, improving navigation and reducing bounce rates
- Contributed to the development of RESTful APIs, enabling seamless third-party platform connections.
- Automated asynchronous coding workflows by integrating AI-based development tools.
- Engineered seamless user flow and implemented real-time data synchronization for AppSumo integration, leveraging Laravel, MySQL, and MariaDB

PROJECTS

SkySense - Flight Analytics Dashboard | *React, JavaScript, Python, Node.js, JSON*

Jan. 2026 – Jan. 2026

- Built an airspace analytics platform simulating 250–1000 flights with real-time multi-aircraft conflict detection.
- Designed a trajectory simulation and conflict-detection engine to identify loss-of-separation events.
- Developed an interactive dashboard to visualize flight paths, conflicts, and resolution status.
- Led development of the core analytics engine and integrated a Python simulation backend with a React frontend via a custom API.

Melanoma Detection Webpage | *Deep Learning, TensorFlow, Streamlit, Git*

Aug. 2025 – Sep. 2025

- Built and trained a CNN in TensorFlow/Keras to classify skin lesions as benign or malignant with 91% accuracy
- Applied preprocessing, augmentation, and class balancing to improve performance on imbalanced medical data
- Developed a Streamlit web app for real-time mole image upload and prediction with confidence scores
- Deployed the project on Streamlit Cloud, making the app publicly accessible and showcasing ML deployment skills

e-Hotels | *Java, PostgreSQL, SQL, JDBC, HTML, CSS, JavaScript*

Jan. 2025 – Apr. 2025

- Designed and implemented a relational database schema supporting hotels, customers, employees, and bookings.
- Developed backend services to support room search, booking, and administrative CRUD operations.
- Implemented complex SQL queries and joins to filter available rooms by capacity, price, city, and hotel chain.
- Integrated a web-based frontend with Java backend services for end-to-end booking workflows.

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript, Kotlin, Ruby, SQL, HTML/CSS

Frameworks & Libraries: React, .NET, Rails, Laravel, JUnit, TensorFlow/Keras, Pandas, NumPy, Streamlit

Databases: PostgreSQL, MySQL, Firebase (Firestore, Realtime Database)

Developer Tools: Git, Docker, Android Studio, IntelliJ IDEA, VS Code, Google Colab

EDUCATION

University of Ottawa

Ottawa, ON

Honours Bachelor of Science in Computer Science (Data Science Option) GPA: 8.55/10

Sep. 2023 – May. 2028

Relevant Coursework: Data Structures & Algorithms, Algorithms, Databases I & II, Operating Systems, Computer Architecture, Programming Paradigms, Probability & Statistics, Data Communication & Networking